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Anoop E G,
A P J Abdul Kalam Technological University, India

*CORRESPONDENCE
Anwesha Mukherjee,
anweshamukherjee2011@gmail.com

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Ensemble learning-based soil moisture monitoring and sustainable irrigation management for Internet of Agricultural Things

Somnath Bera[1], Tanushree Dey[1], Anirbit Sengupta[2], Anwesha Mukherjee[3]* and Debashis De[1]

[1]Department of Computer Science and Engineering, Maulana Abul Kalam Azad University of Technology, Kolkata, West Bengal, India, [2]Department of Computer Science and Engineering-AI, Brainware University, Kolkata, West Bengal, India, [3]Department of Computer Science, Mahishadal Raj College, Mahishadal, West Bengal, India

Introduction: Soil water management is one of the significant domains in smart irrigation. This paper proposes and implements an IoT and edge computing-based soil water management system for sustainable irrigation using ensemble learning.

Methods: An ultralow power sensor node is used to collect real-time soil moisture and soil temperature data. The edge device performs raw data processing and then sends the pre-processed data to the cloud server after adding Gaussian noise to it for data privacy protection. Ensemble learning models are used for analyzing the data inside the cloud.

Results and Discussion: The results are compared to single classifier-based machine learning models and deep learning models. The results demonstrate that the ensemble learning models outperform the single classifier-based machine learning models in terms of prediction accuracy, and the deep learning models in terms of latency and energy consumption. We also perform soil temperature and soil moisture prediction using ensemble learning to deal with the scenario, where either the soil moisture or soil temperature sensor is available, and the results show that the ensemble learning models outperform the baselines. We also analyze the proposed framework's performance for other datasets to further demonstrate its efficacy. The results show that the proposed irrigation decision making framework has ≥99% prediction accuracy for our own dataset and publicly available datasets. Finally, an Android App is designed using which the user can access the data remotely, and can control the irrigation decision making process based on the results.

KEYWORDS
accuracy, latency, smart irrigation, soil moisture, soil temperature

1 Introduction

Soil moisture has a significant impact on the hydrological condition and ecological processes (Ågren et al., 2021). Fresh water is a naturally occurring, ephemeral resource; only 0.01% of the water on the globe is fresh water that may be used for human consumption (Yu et al., 2021). To comprehend water budgets in hydrological systems to minimize the agricultural drought problems, water shortage, and improve crop output globally, it is

crucial to have a thorough grasp of the spatiotemporal patterns of soil moisture (Chaudhary et al., 2022). Moisture is an essential component of soil that influences not only the physical characteristics of the soil but also the dissolution and transport of nutrients in the soil, as well as the activity of microorganisms. The optimum amount of soil moisture is a crucial consideration for preserving good crop growth because different seasons need different soil moisture content for crop yields (Yu et al., 2021). Soil quality monitoring and management play significant roles in sustainable farming (Remya, 2022). Researchers have found that the key determinant of agricultural productivity and hydrological cycle is soil moisture, and accurate soil moisture prediction is crucial for effective soil use and irrigation decision making (Chatterjee et al., 2020). The sustainable irrigation refers to the sensible use of the activities associated with the irrigation, which helps to meet the present survival and welfare needs without affecting the requirements of future generation. The decision making regarding irrigation is a major issue because very high or very low moisture level both can have adverse effect on the plant growth. Manual decision making may not be helpful in such a case. Internet of Things (IoT) has opened a new door (Muangprathub et al., 2019; Fraser Kirwan et al., 2023; Lv, 2022) in smart agriculture. An IoT-based smart irrigation system reduces wastage of water, provides uniform distribution of water, and enhances crop productivity. To further improve the IoT and cloud-based system, edge computing has come (Bera et al., 2023). For privacy-aware data analysis federated learning has come (Dexter Neo and Sheeba, 2026). Federated learning with edge-cloud computing can provide a privacy-aware data analysis through collaborative learning (Soni et al., 2025). Federated learning requires the participating clients or edge nodes to locally train their models and share only model updates for aggregation to generate a global model. However, resource limitation is a crucial issue of edge devices, hence, all the edge nodes may not be able to locally execute a data analysis model.

1.1 Motivations and contributions

The conventional irrigation decision making systems mostly used IoT and cloud computing. However, using cloud servers for raw data storage and processing has a major concern regarding data privacy. Another challenge is to use an energy-efficient device for collecting real-time data. The motivation of the proposed work is to address these issues and provide an intelligent decision making system for sustainable irrigation. To meet these objectives, the contributions of this article are:

- An Edge-Cloud-based Intelligent Irrigation Decision System (ECIID) is proposed.
- An ultra-low power sensor node collects real-time soil temperature and soil moisture data. The edge device performs raw data processing, and sends the pre-processed data to the cloud server after adding Gaussian noise to it for privacy protection during data transmission.
- Ensemble learning models are used for data analysis inside the cloud to build accurate models for irrigation decision making.
- The correlation between soil moisture and soil temperature is observed. Soil moisture and soil temperature prediction models are also developed to make the system able to deal with the scenarios where (i) either soil moisture or soil temperature sensor fails to collect data, and (ii) either the soil moisture or soil temperature data is available.
- An Android App is designed to provide the user (or farmer) the facility of accessing field data anytime, anywhere, and check the decision regarding irrigation.

The rest of the paper is organized as follows. Section 2 briefly discusses the existing works. Section 3 presents the architecture and working model of the proposed system. Section 4 presents the performance analysis of the proposed system. Section 5 concludes the paper with future scope.

2 Related work

This section briefly discusses the existing approaches on soil moisture prediction and irrigation management. Table 1 presents a comparison of the proposed framework with the existing approaches.

IoT and machine learning (ML) were used for precision irrigation in (Manuel Vianny et al., 2022; Tace et al., 2022). An IoT-edge-based two phase smart irrigation framework was developed in (Sahu and Tripathi, 2025). Sensor nodes were used for data collection and ML algorithms were used to predict real time irrigation needs (Sahu and Tripathi, 2025). In (Angelin et al., 2023), a sustainable irrigation management approach was proposed using IoT and transfer learning. In (Yarehalli Chandrappa et al., 2023), spatiotemporal modeling was performed for soil moisture prediction using ML and deep learning (DL) models, such as support vector regression (SVR), linear regression, and long short-term memory (LSTM) network. In (Premkumar and Sigappi, 2022), an IoT-based edge computing framework was proposed for smart irrigation system. The authors used SVR, multiple linear regression (MLR), lasso regression (LR), decision tree regressor (DTR), random forest regressor (RFR), and extreme gradient boosting regression (XGBR), with k-means clustering for smart irrigation management. In (Kaushik and Singh, 2025), artificial intelligence (AI)- and IoT-based smart irrigation system was proposed and LSTM was selected for data analysis. In (Bhanu et al., 2020), an IoT-based smart irrigation system was proposed, and Naive Bayes, k-nearest neighbor (KNN), and support vector machine (SVM) were used for data analysis. In (Medina-García et al., 2025), IEEE 802.15.4 and LoRaWAN were used with sensor network for in-field measurements and large-scale information monitoring for effective irrigation management. In (Khomri et al., 2025), different ML models were compared to predict soil moisture in sugar beet fields in southern Algeria. In (Parvathi Sangeetha et al., 2022), the authors proposed an IoT-based irrigation management system. Various ML methods, such as RFR, XGBR, SVM, and cat boost gradient boosting regression were used to estimate soil moisture with high prediction accuracy in (Ågren et al., 2021; Senanayake et al., 2021). The XGBR was used to build a model using 52 predictor variables from the Sentinel dataset to predict soil moisture in Australia (Nguyen et al., 2022). In (Tripathi et al., 2025), rice yields were predicted by using unmanned aerial vehicle (UAV)-mounted multispectral imagery as input, and ML regressors for data

TABLE 1 Comparison among the proposed and existing approaches on irrigation decision making.

Work	IoT-based system	Edge computing is used	ML-based data analysis	Latency/ Time is measured	Irrigation decision-making	Soil moisture prediction	Soil temperature prediction	Use of GDP
Manuel Vianny et al. (2022)	✓	✗	✓	✓	✓	✗	✗	✗
Sahu and Tripathi (2025)	✓	✓	✓	✓	✓	✗	✗	✗
Tace et al. (2022)	✓	✗	✓	✗	✓	✗	✗	✗
Angelin et al. (2023)	✓	✗	✓	✗	✓	✗	✗	✗
Yarehalli Chandrappa et al. (2023)	✓	✗	✓	✗	✓	✗	✗	✗
Premkumar and Sigappi (2022)	✓	✓	✓	✓	✗	✗	✗	✗
Kaushik and Singh (2025)	✓	✓	✓	✓	✓	✗	✗	✗
Bhanu et al. (2020)	✓	✗	✓	✗	✓	✗	✗	✗
ECIID (proposed)	✓	✓	✓	✓	✓	✓	✓	✓

analysis. In (Dubois et al., 2021), supervised learning was used to address the soil water potential problem by automatically building lighter models with the best features.

Though several approaches exist on soil moisture prediction and irrigation decision making, most of the works used various parameters for estimation and prediction, which make the framework expensive. Further, using a remote cloud for raw data processing, storage, and analysis compromises data privacy and security. The proposed work addresses these issues by proposing an intelligent irrigation decision making framework based on IoT with ultra-low power sensor node, edge-cloud computing, ensemble learning, and Gaussian differential privacy.

2.1 Limitations of the existing approaches

Most of the existing approaches use various sensors and parameters for irrigation decision making, and use cloud servers for raw data processing, storage, and analysis. In the developing countries, the use of various types of sensors for data collection may not be economically feasible from the farmers' perspective. Moreover, remote cloud usage for raw data processing is a major concern with respect to data privacy. Another major concern is the power consumption of the sensor node used for data collection. In our paper, we have provided a solution towards these issues by introducing an IoT and edge-cloud-based system named as ECIID that (i) collects real-time soil moisture and soil temperature data using ultra-low power sensor, (ii) performs raw data processing inside the edge device, adds Gaussian noise to the pre-processed data, and sends the Gaussian differential privacy (GDP)-based data to the cloud, and (iii) uses ensemble learning for data analysis inside the cloud for irrigation decision making. For secure data transmission and protection of privacy, the GDP is used in ECIID. Table 1 shows that the proposed framework ECIID is unique and has better features compared to the existing approaches.

We have also compared the proposed ultra-low power data collection device used in ECIID with the existing technologies [MICA2DOT (https://www.btnode.ethz.ch/pub/files/btnode_rev3.22_productbrief.pdf) and Prototype Model (Bin Poyen et al., 2020)] in Table 2. The block diagram of the sensor node is presented in Figure 1. As observed from Table 2, edge device and cloud were not used in the existing technologies. In our system, the cloud is used for providing the facility of storing a large volume of data as well as the user can access their data anytime, anywhere. The Li-Ion battery is recharged by a solar panel and using TP4056 charger. The use of ultra-low power sensor node provides an eco-friendly data collection in ECIID. After collecting the real-time soil moisture and soil temperature data, the raw data processing is performed inside the edge device. Before data transmission to the cloud, Gaussian noise is added to the pre-processed data for privacy protection by maintaining the data pattern. For data analysis ensemble learning is used, and the decision regarding irrigation is taken. Though, our device is able to collect both the soil moisture

and soil temperature data, we have also considered the scenario, where either soil temperature or soil moisture sensor fails, or either the soil moisture data or soil temperature data is available. In such a case, the soil temperature (if soil moisture data is available) or soil moisture (if soil temperature data is available) is predicted, so that based on the collected and predicted data, the irrigation decision can be made.

3 ECIID: the proposed framework

The three-tier IoT-edge-cloud framework used for soil moisture and soil temperature data collection, and analysis is briefly described in this section, followed by the ensemble learning models used for data analysis.

3.1 Architecture of ECIID

The three-tier architecture used in our system is presented in Figure 2, and is described as follows:

- Tier 1: contains the ultra-low power sensor node that contains soil temperature and soil moisture sensors, and the microcontroller unit. A sensor node is mathematically represented by Equation 1 as a four tuple containing identity (s_{id}), configuration (s_{con}), object type (s_{obj}), and mode (s_{mode}), given as,

$$s = < s_{id}, s_{con}, s_{obj}, s_{mode} > \quad (1)$$

The mode of the sensor node can be idle or active. The sensor node is deployed in the agricultural land of consideration, and connected with the edge device. The sensor node collects the soil moisture and soil temperature data of the respective land, and then the data is stored inside the microcontroller. The data is transmitted to the edge device in tier 2 at a regular interval. In our experiment, the data is sent to the edge device at 30 min interval.

- Tier 2: The edge devices are present in tier 2, which perform the transformation of raw data to compatible format required for analysis. The edge device is mathematically defined by Equation 2 as a three tuple containing its identity (e_{id}), configuration (e_{con}), and mode (e_{mode}), given as,

$$e = < e_{id}, e_{con}, e_{mode} > \quad (2)$$

In our experiment, Raspberry Pi is used as the edge device that converts the raw data into a csv file, pre-processes it, adds Gaussian noise to the pre-processed data, and sends the GDP-based data to the cloud server along with the information regarding the soil type i.e., sandy, loamy, clayey, etc. The edge device sends the data to the cloud at 24 h interval, i.e., once per day, in our experiment. For secure data transmission, GDP (Dong et al., 2022) is used to add Gaussian noise to the dataset for protecting data without affecting the underlying pattern of the dataset. Instead of encryption, we have used GDP in ECIID for a lightweight security and privacy protection.

- Tier 3: Tier 3 contains the cloud server that stores and analyzes the received data. The cloud server instance is mathematically defined by Equation 3 as a two tuple containing instance identity (c_{id}) and processing ability (c_{proc}), given as,

$$c = < c_{id}, c_{proc} > \quad (3)$$

In the experiment, the correlation between the soil temperature and soil moisture is determined. For data analysis, ensemble learning models are used, described in detail in Section 3.2.

3.2 Data analysis using ensemble learning

In ECIID, we have performed data analysis using ensemble learning. The advantages of ensemble learning over the traditional single classifier-based ML are better prediction accuracy, robustness, generalization, reduced overfitting, etc. Ensemble learning is of three types: bagging, boosting, and stacking, briefly described as follows:

- *Bagging:* In bagging, models are independently trained on different subsets of the training data selected randomly, and the results are combined to generate the final result. Usually, for classification, voting is used, and for regression, averaging is used. In Figure 3a, bagging using random forest (RF) is pictorially presented. The dataset is divided into N subsets, and training is performed on the randomly selected subsets using decision tree denoted as DT. Finally, the results are combined using averaging or voting to get the final result.
- *Boosting:* In boosting, multiple weak learners are combined to build a strong learner. The models are sequentially trained with an objective for each model to rectify errors of the previous model until the correct prediction is performed for the entire dataset. Figure 3b presents the boosting process using extreme gradient boosting (XGBoost). The dataset is divided into N subsets, and training is performed on the subsets to get multiple week learners (WL) using decision tree. These weak learners are combined to get the final result using model weighs.
- *Stacking:* Different types of models are trained, and their predictions are considered as inputs for the final model, referred to as the meta model. The meta model learns the best combination of the predictions of the base models for better prediction. Figure 3c presents the stacking process using RF and XGBoost as the base models, and logistic regression as the meta model.

In ECIID, data analysis is performed using bagging with RF, boosting with XGBoost, and stacking with RF and XGBoost.

3.3 Latency in ECIID

The total latency in ECIID is measured as the sum of the data transmission latency, raw data processing time including data pre-processing, Gaussian noise addition time, training time, and prediction time for the test dataset. The parameters used for latency calculation for the proposed system ECIID are defined in Table 3.

The latency for data transmission from the sensor node to the edge device is determined using Equation 4 as follows:

$$L_{se} = (1 + \phi_{se}) * (D_{se}/R_{se}) \quad (4)$$

TABLE 2 Comparison of the ultra-low power sensor node used in our system with the existing technologies.

Sensor	Data collection device in ECIID	MICA2DOT	Prototype model (Bin Poyen et al., 2020)
Processor	MSP430G2553	ATmega128L	ATmega32 Nano
Communication Module	M95 GSM MODEM	CC1000 radio transceiver	SIM800
Sensors	Soil Moisture, Soil Temperature	Temperature and battery monitor	Temperature, pH, Hall, humidity, etc.
GPIO pins	27	18 solder-less expansion pins	No GPIO pin open for user
CLOUD STORAGE	✓	✗	✗
Processor in Bit	16-Bit	8-Bit	8-Bit
Power used	Solar and Li-Ion battery	Li-Ion battery	Li-Ion battery
Edge device attached	✓	✗	✗

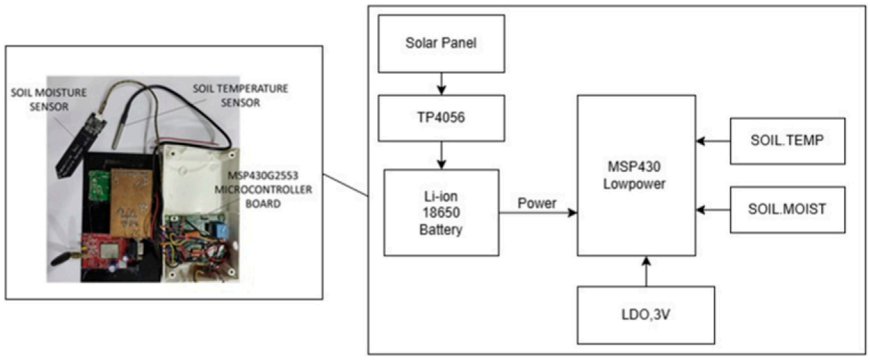


FIGURE 1
Block diagram of the proposed sensor node used for data collection.

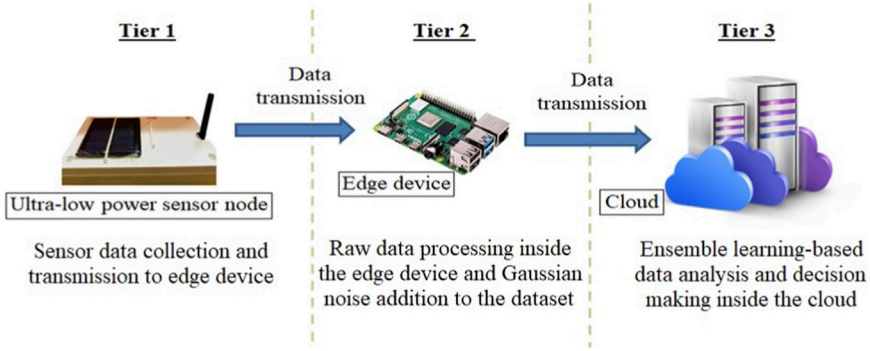


FIGURE 2
Three-tier architecture of proposed system.

The latency for data transmission from edge device to the cloud is determined using Equation 5 as follows:

$$L_{ec} = (1 + \phi_{ec}) * (D_{ec}/R_{ec}) \quad (5)$$

The total latency for data transmission is calculated using Equation 6 as follows:

$$L_{tr} = L_{se} + L_{ec} \quad (6)$$

The total latency is calculated using Equation 7, as the sum of the data transmission latency, raw data processing time including data pre-processing, Gaussian noise addition time, training time, and prediction time, as follows:

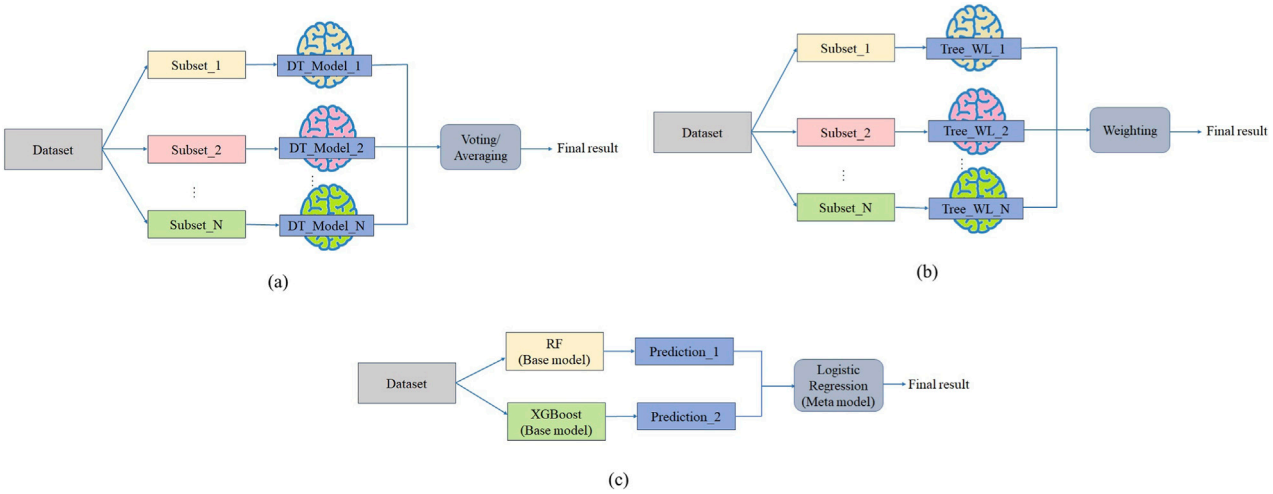


FIGURE 3
Types of ensemble learning. **(a)** Bagging, **(b)** Boosting, **(c)** Stacking.

TABLE 3 Parameters used for latency calculation.

Parameter	Definition
D_{se}	Amount of data transmitted from sensor node to edge device
R_{se}	Data transmission rate from sensor node to edge device
ϕ_{se}	Link failure rate from sensor node to edge device
L_{pre}	Raw data processing including data pre-processing time
L_{en}	Latency in adding Gaussian noise to dataset
D_{ec}	Data amount transmitted from edge device to cloud
R_{ec}	Data transmission rate from edge device to cloud
ϕ_{ec}	Link failure rate from edge device to cloud
L_{train}	Training time
L_{pr}	Prediction time

$$L_{tot} = L_{tr} + L_{pre} + L_{en} + L_{train} + L_{pr} \tag{7}$$

The latency for ECIID is measured from the experiment in Section 4.

4 Performance analysis of ECIID

This section presents the implementation of the proposed system ECIID and evaluates its performance in terms of prediction accuracy, latency, and energy consumption.

4.1 Implementation of ECIID

We have used an ultra-low power sensor node containing soil moisture sensor, soil temperature sensor, and the microcontroller unit. DS18B20 soil temperature sensor and a capacitive type soil moisture sensor are used. The ultra-low power sensor node is deployed inside the agricultural land under consideration. The sensors have used I2C protocol to connect to the ultra-low power MSP430G2553 controller and fetch real-time data from the field. The sensor node is powered using Li-Ion battery which is charged using a solar panel. The use of solar panel makes the data collection process of the ECIID framework sustainable and energy-efficient. The sensors are inserted 10 cm into the surface of the soil to collect real-time data of soil temperature and soil moisture. The sensor node is connected to the Internet through an M95 GSM MODEM. The acquired sensor data is saved in an EPROM of the microcontroller, and then is sent to the linked edge device at 30 min interval using Wi-Fi technology. The Raspberry Pi is employed as the edge device. The Raspberry Pi's 16 GB of RAM is used to transform the microcontroller's raw data into csv format. In our experiment, a python script is developed, run, and then a csv file is created, along with the date, time, and the received sensor data. The data pre-processing is performed to replace the missing values by the mean value of the respective data attribute, and remove the outliers, and then generate a cleaned dataset. Then, Gaussian noise is added to the dataset, and the csv file containing the GDP-based dataset is uploaded to the cloud every 24 h. The Raspberry Pi also sends information about the soil type (sandy, loamy, clayey, etc.) during uploading the csv file. In our work, Gaussian noise is added to the dataset before transmission to the cloud to protect data privacy and enhance security. Based on the privacy budget ($Priv_{noise}$) and the probability ($Prob_{break}$) of breaking differential privacy to obtain the original data, the Gaussian noise addition is expressed in Equation 8:

$$\mathcal{G}(Data) = \mathcal{F}(Data) + \mathcal{Y} \tag{8}$$

where $\mathcal{F}$ denotes the function, $\mathcal{Y}$ denotes the Gaussian noise, $\mathcal{Y} \sim Noise(0, \sigma^2)$, $\Delta\mathcal{F}$ denotes the sensitivity of the function $\mathcal{F}$, and $\sigma \geq \sqrt{2\ln\frac{1.25}{Prob_{break}}} \cdot \frac{\Delta\mathcal{F}}{Priv_{noise}}$. Though GDP is a popular approach, there are some critical concerns: (i) the addition of continuous Gaussian noise can distort the underlying data distribution and the noisy outputs can fall outside the valid data ranges, (ii) there is also a probability of finite precision and implementation vulnerabilities due to the addition of continuous and unbounded Gaussian noise, and (iii) the addition of Gaussian noise to the data can cause a

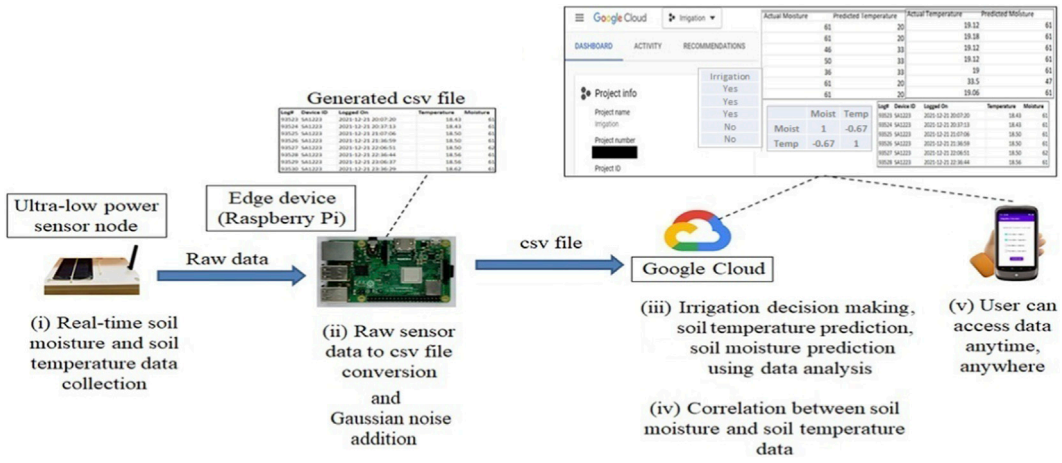


FIGURE 4
Implementation of the proposed system (ECIID).

reduction in model accuracy. Hence, a privacy-accuracy trade-off should be maintained. In the present work, a Gaussian noise within the scale (0, 1) is added to maintain the underlying data distributions of the original dataset. For data analysis using ensemble learning models, Google Cloud Platform (GCP) is used. To link the Raspberry Pi with the Google Cloud, we use NodeJS client that transfers the data through Message Queuing Telemetry Transport (MQTT) protocol. Figure 4 pictorially presents the implementation of the proposed system. The data collection is performed for 90 days from three different regions with different soil types (loamy, sandy, and clayey).

4.2 Experimental results

To evaluate the performance of the proposed system, we consider accuracy, precision, recall, and F1-score, as the performance metrics in data analysis, and then determine the latency and energy consumption of the proposed framework. The accuracy ($\mathcal{A}$), precision ($\mathcal{P}$), recall ($\mathcal{R}$), and F1-score ($\mathcal{F}$) are calculated as follows (Bera et al., 2023):

$$\mathcal{A} = \frac{(\tau + \gamma)}{(\tau + \delta + \gamma + \nu)},$$

$$\mathcal{P} = \frac{\tau}{(\tau + \delta)},$$

$$\mathcal{R} = \frac{\tau}{(\tau + \nu)},$$

and

$$\mathcal{F} = \frac{2 * \mathcal{P} * \mathcal{R}}{(\mathcal{P} + \mathcal{R})}$$

where τ, δ, γ, and ν represent true positive, false positive, true negative, and false negative values, respectively.

4.2.1 Prediction accuracy of irrigation decision making

Firstly, the decision regarding irrigation is made after analyzing the soil moisture and soil temperature data contained in the csv file, using ensemble learning models. There are two labels or categories for irrigation decision: yes (1) and no (0). For labeling in our own dataset, we have considered the soil moisture and soil temperature values based on the soil type. In our experiment, we have considered loamy soil, sandy soil, and clayey soil. For loamy soil, (i) if soil temperature is $> 30^0$C and soil moisture is $\leq 40\%$, the irrigation label is set to yes (1), (ii) otherwise the irrigation label is set to no (0).

TABLE 4 Prediction accuracy of the ensemble learning models using k-fold cross validation.

Value of k	Method	Accuracy
k = 1	Boosting (XGBoost)	98.7%
	Bagging (RF)	98.6%
	Stacking (RF, XGBoost)	98.6%
k = 2	Boosting (XGBoost)	99%
	Bagging (RF)	98.8%
	Stacking (RF, XGBoost)	98.8%
k = 3	Boosting (XGBoost)	99.05%
	Bagging (RF)	99.1%
	Stacking (RF, XGBoost)	99%
k = 4	Boosting (XGBoost)	99.1%
	Bagging (RF)	99.2%
	Stacking (RF, XGBoost)	99.2%
k = 5	Boosting (XGBoost)	99.15%
	Bagging (RF)	99.3%
	Stacking (RF, XGBoost)	99.4%
Mean accuracy	Boosting (XGBoost)	99%
	Bagging (RF)	99%
	Stacking (RF, XGBoost)	99%

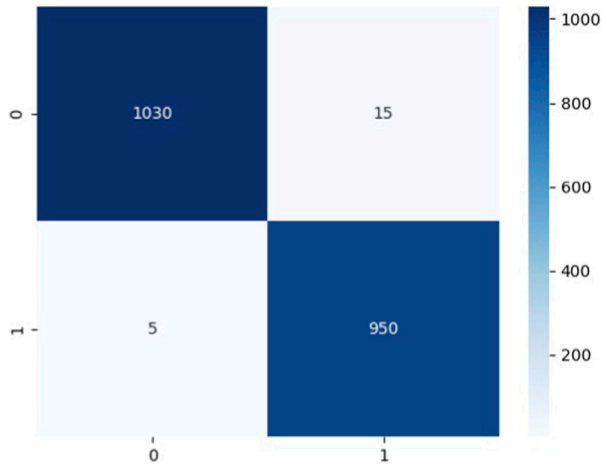


FIGURE 5
Confusion matrix for boosting (XGBoost).

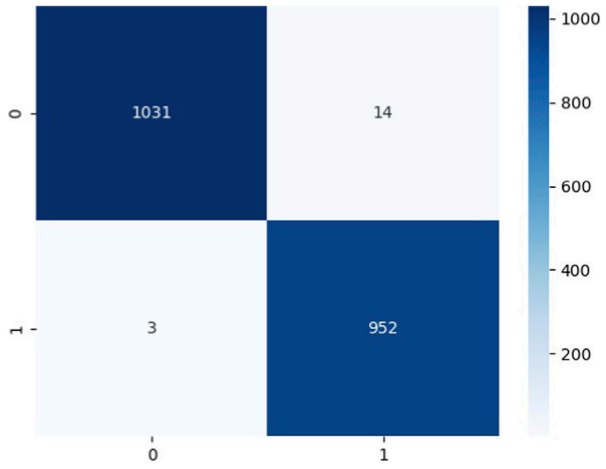


FIGURE 7
Confusion matrix for stacking (RF, XGBoost).

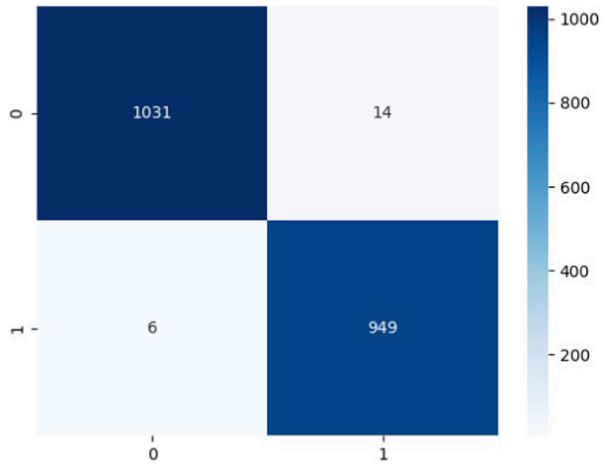


FIGURE 6
Confusion matrix for bagging (RF).

For sandy soil, (i) if soil temperature is 15^0C to 30^0C or above 30^0C and soil moisture is $\leq 50\%$, the irrigation label is set to yes (1), (ii) otherwise the irrigation label is set to no (0). For clayey soil, (i) if soil temperature is above 35^0C and soil moisture is $\leq 40\%$, the irrigation label is set to yes (1), (ii) otherwise the irrigation label is set to no (0). The dataset used for analysis has 10000 samples, and it is split into 80% and 20% ratio to generate a train dataset containing 8000 samples, and a test dataset containing 2000 samples. The soil temperature and soil moisture data are analyzed using ensemble learning models, and a decision is made whether irrigation is required or not. We have performed k-fold cross-validation to overcome the overfitting issue and observed that a high value of mean accuracy level is obtained. The value of "k" is set to 5. The prediction accuracies of the ensemble learning models for the k-fold cross validation are presented in Table 4. We observe that the highest accuracy is achieved for k = 5, and mean accuracies of all ensemble learning models are 99%. The confusion matrices for the validation dataset for boosting, bagging, and stacking are presented in Figures 5, 6, 7, respectively. Table 5 compares the mean prediction accuracies of the ensemble learning models with single classifier-based ML models like KNN and decision tree (DT), and DL models like LSTM and gated recurrent unit (GRU), considered as the baselines. The total latency and energy consumptions are also presented in the table. The energy consumption of the decision making framework is obtained as the product of the total latency and the power consumption of all participating nodes including sensor node, edge device, and cloud server per unit time. We observe from the results that the ensemble learning models have achieved 99% accuracy, precision, recall, and F1-score, and outperformed the single classifier-based ML models. The results show that among the ensemble learning models, (i) stacking has higher latency and energy consumption than other ensemble learning models and the single classifier-based ML models, (ii) bagging has slightly higher latency and energy consumption than the single classifier-based ML models and boosting, and (iii) boosting has approximately the same latency and energy consumption as the single classifier-based ML models, and lower than other ensemble learning models. Though the DL models have 99% prediction accuracy, they have higher latency and energy consumption due to high computational complexity compared to the ensemble learning models and single classifier-based ML models. We have also performed a comparison of the prediction accuracy of the ensemble learning models for the original dataset and the GDP-based dataset in Table 6. We observe that the prediction accuracy for the original dataset (without adding Gaussian noise) are 99.5% for boosting, bagging, and stacking. We observe from the analysis that the drop in accuracy due to the use of GDP is very less for our own dataset. We also observe that the total latency and energy consumption are slightly high in case of GDP-based data analysis, because a very small amount of time consumption and respective energy consumption are involved for adding Gaussian noise to the dataset for data privacy protection.

4.2.2 Correlation between soil moisture and soil temperature

To determine the statistical relationship between two variables, *correlation* is used. The correlation coefficient is a

TABLE 5 Experimental results of irrigation decision making for our dataset.

Method	Accuracy	Precision	Recall	F1-score	Total latency (s)	Energy (J)
KNN	98%	98%	98%	98%	0.45	315
DT	98%	98%	98%	98%	0.45	315
LSTM	99%	99%	99%	99%	12.38	8700
GRU	99%	99%	99%	99%	10.33	7240
Boosting (XGBoost)	99%	99%	99%	99%	0.45	315
Bagging (RF)	99%	99%	99%	99%	1.77	1250
Stacking (RF, XGBoost)	99%	99%	99%	99%	9.21	6450

TABLE 6 Comparison of performance of the ensemble learning models for the original and GDP-based datasets.

Performance metric	Model	GDP-based dataset	Original dataset
Accuracy	Boosting (XGBoost)	99%	99.5%
	Bagging (RF)	99%	99.5%
	Stacking (RF, XGBoost)	99%	99.5%
Precision	Boosting (XGBoost)	99%	100%
	Bagging (RF)	99%	100%
	Stacking (RF, XGBoost)	99%	100%
Recall	Boosting (XGBoost)	99%	99%
	Bagging (RF)	99%	99%
	Stacking (RF, XGBoost)	99%	99%
F1-score	Boosting (XGBoost)	99%	99.5%
	Bagging (RF)	99%	99.5%
	Stacking (RF, XGBoost)	99%	99.5%
Total latency (s)	Boosting (XGBoost)	0.45	0.42
	Bagging (RF)	1.77	1.7
	Stacking (RF, XGBoost)	9.21	9.15
Energy	Boosting (XGBoost)	315	310
	Bagging (RF)	1250	1225
	Stacking (RF, XGBoost)	6450	6420

number derived from the data that expresses how strongly two variables are linearly related. The coefficients have values ranging from −1 (perfect inverse correlation) to 1 (perfect direct correlation), with zero denoting no association. Figure 8 shows the correlation between the original soil moisture and soil temperature data, and Figure 9 shows the correlation between the GDP-based soil moisture and soil temperature data (after adding Gaussian noise to the data). The results show that the soil moisture and soil temperature are closely related to each other for the original dataset as well as for the GDP-based dataset. The results also show that the correlation coefficient for the original data closely matches with the GDP-based data. The results show that the underlying data distributions of the original dataset is maintained after adding Gaussian noise within a scale (0, 1) for privacy protection.

4.2.3 Prediction of soil moisture and soil temperature

There is a possibility of sensor failure in agricultural fields or a situation may arise where either the soil moisture or soil temperature sensor is available. If either the soil moisture sensor or soil temperature sensor is available, then the soil temperature values (from moisture data) or moisture values (from temperature

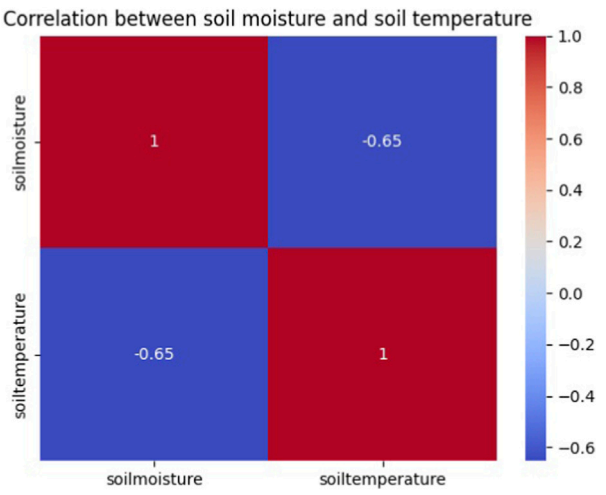


FIGURE 8
Correlation between soil moisture and soil temperature in original dataset.

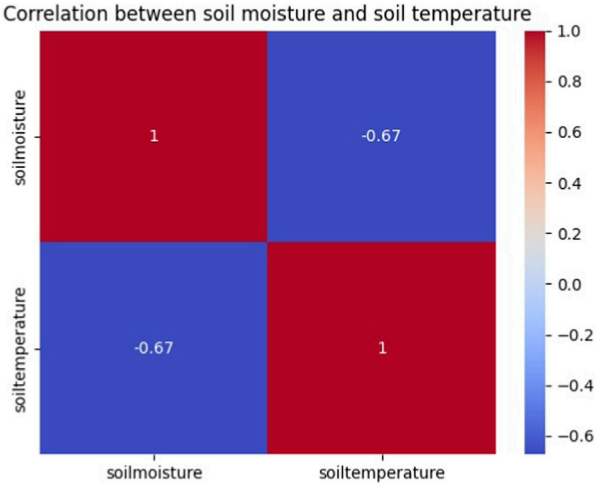


FIGURE 9
Correlation between soil moisture and soil temperature in GDP-based dataset.

data) can be predicted using a trained model. In the train dataset, both soil moisture and soil temperature data are available. The test dataset contains either soil moisture or soil temperature data. We have considered this scenario assuming either the soil moisture sensor or the soil temperature sensor is working, and the decision regarding irrigation needs to be made. For this prediction, again we use ensemble learning-based regression models: bagging using RF regression, boosting using XGBoost regression, and stacking with RF and XGBoost regression models. The mean absolute error (MAE), root mean squared error (RMSE), and the coefficient of determination (R^2) are measured as the evaluation metrics. The MAE ($\mathcal{E}_1$), RMSE ($\mathcal{E}_2$), and R^2 ($\mathcal{E}_3$) are calculated as follows:

TABLE 7 Experimental results of soil temperature prediction.

Method	MAE	RMSE	R^2
KNN	0.985	1.22	0.22
DT	1.05	1.35	0.11
XGBoost (Boosting)	0.88	1.13	0.33
RF (Bagging)	0.98	1.2	0.24
Stacking (RF, XGBoost)	0.81	1.01	0.48

TABLE 8 Experimental results of soil moisture prediction.

Method	MAE	RMSE	R^2
KNN	8.83	11.47	0.29
DT	9.68	12.82	0.073
XGBoost (Boosting)	8.64	11.2	0.3
RF (Bagging)	8.43	11.03	0.3
Stacking (RF, XGBoost)	7.91	10.3	0.45

$$\mathcal{E}_1 = \frac{1}{\mathcal{K}} \sum_{i=1}^{\mathcal{K}} |\alpha_i - \beta_i|,$$

$$\mathcal{E}_2 = \sqrt{\frac{\sum_{i=1}^{\mathcal{K}} (\alpha_i - \beta_i)^2}{\mathcal{K}}},$$

and

$$\mathcal{E}_3 = 1 - \frac{\sum_{i=1}^{\mathcal{K}} (\alpha_i - \beta_i)^2}{\sum_{i=1}^{\mathcal{K}} (\alpha_i - \mu)^2},$$

where $\mathcal{K}$ is the number of observations, α denotes the actual value, β denotes the predicted value, and μ denotes the mean value.

Tables 7, 8 present the experimental results of soil temperature and soil moisture prediction, respectively. We observe from Tables Tables 7, 8 that RF (bagging), XGBoost (boosting), and stacking using RF and XGBoost, have achieved lower MAE and RMSE values, and better R^2 score, compared to KNN and DT, considered as the baselines. We also observe that from soil moisture data, soil temperature can be predicted more accurately compared to the prediction of soil moisture from soil temperature data.

4.2.4 Performance analysis for other datasets

The performance of the decision making model for other datasets (Dataset[1] (https://data.mendeley.com/datasets/fpdwmm7nrb/1) and Dataset[2] (https://www.kaggle.com/datasets/harshilpatel355/autoirrigationdata)) is also evaluated to check the robustness and scalability of the proposed approach. In Dataset[1] there are four columns: the first three columns denote the input features (soil moisture, air temperature, and air humidity), and the last column denotes the output label (motor on/off). There are 3000 samples in the cleaned dataset obtained after pre-processing. On this dataset, we have applied GDP (Gaussian noise within scale (0, 1)). Then, the dataset is split into 80% and 20% ratio to generate

TABLE 9 Experimental results of irrigation decision making for other datasets.

Dataset	Method	Accuracy	Precision	Recall	F1-score	Latency (s)	Energy (J)
Dataset[1]	KNN	0.97	0.97	0.97	0.97	0.16	192
	DT	0.98	0.98	0.98	0.98	0.16	192
	LSTM	0.99	0.99	0.99	0.99	4.25	5100
	GRU	0.99	0.99	0.99	0.99	3.35	4050
	XGBoost (Boosting)	0.99	0.99	0.99	0.99	0.16	192
	RF (Bagging)	0.99	0.99	0.99	0.99	0.65	780
	Stacking (RF, XGBoost)	0.992	0.99	0.99	0.99	3.05	3650
Dataset[2]	KNN	0.88	0.9	0.88	0.88	0.015	18.2
	DT	0.99	0.99	0.99	0.99	0.015	18.2
	LSTM	0.995	0.995	0.995	0.995	0.45	540
	GRU	0.995	0.995	0.995	0.995	0.34	410
	XGBoost (Boosting)	0.995	0.995	0.995	0.995	0.015	18.2
	RF (Bagging)	0.995	0.995	0.995	0.995	0.068	85
	Stacking (RF, XGBoost)	0.995	0.995	0.995	0.995	0.31	370

TABLE 10 Comparison of proposed and existing irrigation decision making systems.

Work	Classifier	Accuracy
Sahu and Tripathi (2025)	XGBoost	99.96%
Tace et al. (2022)	KNN (best), Neural network, SVM, Naive Bayes, Logistic Regression	98.3%
Yarehalli Chandrappa et al. (2023)	Linear regression (best), SVR, LSTM	98.33%
Premkumar and Sigappi (2022)	SVR + k-means, MLR + k-means, LR + k-means, DTR + k-means, RFR + k-means, and XGBR + k-means (best)	97%
Kaushik and Singh (2025)	LSTM	92.5%
Bhanu et al. (2020)	SVM (best), KNN, Naive Bayes	87.5%
ECIID (proposed)	Ensemble learning using RF, XGBoost	≥ 99%

the training and validation datasets. Dataset[2] contains three columns as input features (crop, moisture, and temperature), and one column for the output label (pump on/off). There are 200 samples in the cleaned dataset obtained after pre-processing, and to the samples Gaussian noise is added within scale (0, 1). After that these samples are split into 80% and 20% ratio to generate the training and validation datasets. In Table 9, we compare the accuracy, precision, recall, and F1-score of the ensemble learning models with the single classifiers like KNN and DT, and DL models like LSTM and GRU. The results show that the ensemble learning models XGBoost, RF, and stacking have achieved better prediction accuracy than the single classifier-based ML models, for both dataset[2] and dataset[3]. The data transmission latency, training time, and validation time are added to measure the total latency. The energy consumption is measured as the product of the latency and the power consumption of the edge node and the cloud server per unit time. However, the latency and energy consumption are higher for the stacking process compared to other ensemble learning models. The bagging algorithm (RF) has slightly higher latency and energy consumption than the single classifier-based ML models. The DL models though achieved high prediction accuracy, the latency and energy consumption are higher than the ensemble learning models. The boosting algorithm (XGBoost) has high prediction accuracy with low latency and low energy consumption for both the datasets.

4.2.5 Comparison with existing irrigation decision making systems

Table 10 shows a comparative study of the proposed and existing irrigation decision making systems. In (Sahu and Tripathi, 2025), XGBoost was used, and the achieved accuracy was 99.96%. In (Tace et al., 2022), the authors used KNN, neural network, SVM, Naive Bayes, and logistic regression, and among them KNN achieved the highest accuracy of 98.3%. In (Yarehalli Chandrappa et al., 2023), LSTM, SVR, and linear regression were used, and among them linear regression achieved the highest accuracy of 98.33%. In (Premkumar and Sigappi, 2022), SVR, MLR, LR, DTR, RFR, and



FIGURE 10
User login page of the proposed Android App.

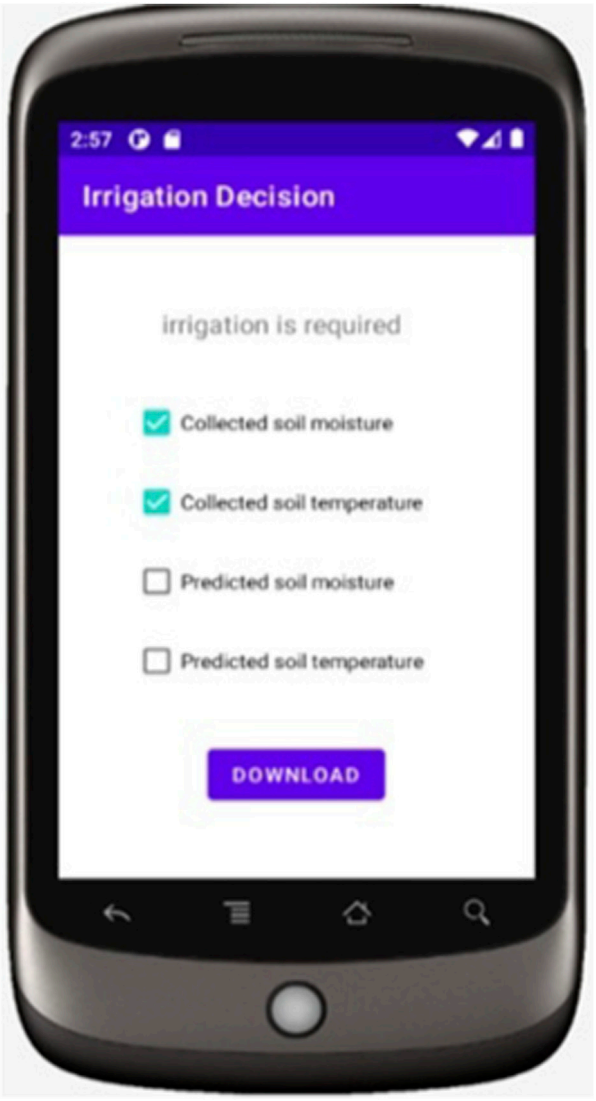


FIGURE 11
View irrigation decision using the proposed Android App.

XGBR, were used with k-means, and among them XGBR with k-means achieved the highest prediction accuracy of 97%. In (Kaushik and Singh, 2025), LSTM achieved an accuracy of 92.5%. In (Bhanu et al., 2020), the authors used Naive Bayes, KNN, and SVM, and among them SVM achieved the highest accuracy of 87.5%. Our proposed work has achieved $\geq 99\%$ prediction accuracy considering our dataset and publicly available datasets. As observed from the table, the proposed approach achieves higher accuracy than most of the existing systems. Further, the use of GDP provides privacy protection in ECIID, which makes it unique compared to the existing schemes.

4.3 Android App for irrigation decision

We propose an Android App to provide the user an opportunity to access the soil temperature and/or soil moisture data of his/her agricultural field anytime, anywhere, and view irrigation decision using his/her mobile device.

At first the user performs login by providing proper user credentials (user id and password), as presented in Figure 10. Then, the user can view the decision regarding irrigation as shown in Figure 11. The user also has the opportunity to download the data in the form of csv file by clicking on the download button. As in our approach, we are providing an opportunity to predict the soil moisture from soil temperature data and *vice versa*, if anyone has only the soil moisture or soil temperature sensor, we are providing the option to download the collected as well as predicted data in our app. The user can also view the irrigation decision based on data analysis, so that the user can accordingly control the irrigation management process remotely. However, as GDP is applied before data transmission from the edge device to the cloud, the accessed dataset maintains the pattern of the original dataset but not the actual values. This is the limitation of the designed app.

The results show that the proposed approach provides an accurate, latency-aware, and energy-efficient irrigation decision making. Now, we will discuss how the proposed ECIID framework can address several challenges such as scalability,

communication reliability, and maintenance requirements in the practical deployment. As the data analysis is performed inside the cloud, and cloud computing is a scalable computing paradigm, numerous edge devices can be connected to the cloud for a large-scale data processing. However, due to cloud-based analysis, a good network connectivity is required during pre-processed data transmission from the edge device to the cloud. As raw data processing is happened inside the edge device and GDP is used on the pre-processed data before transmission to the cloud, data privacy is protected and security is enhanced. As the proposed framework uses IoT-edge-cloud computing, maintenance is also a vital issue. Periodic sensor calibration, regular update of the edge device including its operating system, and automated cloud orchestration will be performed for maintenance of the proposed framework. To validate further the proposed framework's performance, we have conducted A/B testing (https://github.com/AnuTuli/ECIID/blob/main/AB_test_ECIID.pdf). The results also show that for our dataset, all of the ensemble learning models (boosting (XGBoost), bagging (RF), and stacking using RF and XGBoost) achieved Z-score of above 2.58 and P-value of 0, which indicate that the proposed framework is robust and it has a high confidence level.

5 Conclusion and future work

IoT and edge computing have become two principal components of designing smart systems. In this paper, we have proposed and implemented an IoT and edge-cloud-based intelligent system for irrigation decision making. In the proposed system, we have used an ultra-low power sensor node for collecting real-time soil moisture and soil temperature data. The edge device performs raw data processing on the collected data, adds Gaussian noise to the pre-processed data, and then sends the data to the cloud. The cloud stores the received GDP-based data and performs analysis on the data using ensemble learning models. The performances of the ensemble learning models are compared with the single classifier-based ML models like KNN and DT, and DL models like LSTM and GRU, in terms of accuracy, precision, recall, and F1-score. From the experiment, we have also measured the latency including data transmission, raw data processing, Gaussian noise addition to the pre-processed data, and analysis. The results show that (i) the ensemble learning models have higher prediction accuracy, precision, recall, and F1-score than the single classifier-based ML models, and (ii) the ensemble learning models have achieved the high prediction accuracy like the DL models but with much lower latency and energy consumption. The results also show that the ensemble learning models have achieved $\geq 99\%$ accuracy for our dataset and publicly available datasets. To deal with the scenarios where only the soil moisture or soil temperature sensor is available, soil temperature and soil moisture prediction models are also proposed. The results show that the ensemble learning models have outperformed the single classifier-based ML models in soil moisture and soil temperature predictions.

In an agricultural region, both the soil moisture and soil temperature sensors can fail, or extreme weather conditions may occur. In the future, we would like to propose a soil monitoring and an irrigation management approach for such situations. In the future, we also wish to propose a split learning-based irrigation decision making model for precision agriculture.

Data availability statement

The raw data supporting the conclusions of this article will be made available by the authors, without undue reservation.

Author contributions

SB: Conceptualization, Formal Analysis, Investigation, Methodology, Validation, Writing – original draft. TD: Conceptualization, Formal Analysis, Investigation, Methodology, Validation, Writing – original draft. AS: Data curation, Formal Analysis, Investigation, Methodology, Validation, Writing – original draft. AM: Formal Analysis, Methodology, Supervision, Validation, Writing – original draft, Writing – review and editing. DD: Resources, Supervision, Validation, Writing – review and editing.

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